buildandteach

PYSPARK QUERY · PART 2 QUICK REFERENCE

groupBy · aggregations · windows

Change the grain on purpose. Rank with windows while keeping detail.
Free Edition: serverless + UC. Read-only. Same 8 fictional orders.

1. Valid input gate

```
valid = orders.filter(F.col("quantity").isNotNull() &
                      F.col("unit_price").isNotNull())
# Actual: 6 valid rows · revenue 142.50
```

Reconcile every later total back to 142.50.

2. Region × category aggregation

```
by_rc = (valid.groupBy("region","category")
  .agg(F.count("*").alias("order_count"),
       F.sum("order_total").cast("decimal(12,2)").alias("revenue"))
  .orderBy("region","category"))
```

Actual 5 buckets: E/Books 12 · N/Books 20 · N/Toys 60 · S/Toys 25 · W/Books 25.50.

3. Region rollup

```
by_region = (valid.groupBy("region")
  .agg(F.count("*").alias("order_count"),
       F.sum("order_total").cast("decimal(12,2)").alias("revenue")))
```

Actual: EAST 12 · NORTH 80 (3) · SOUTH 25 · WEST 25.50.

Windows + Spark WARN teaching point

4. Rank + running revenue

```
w_rank = Window.partitionBy("category")
  .orderBy(F.col("order_total").desc(), F.col("order_id").desc())
w_run = (Window.orderBy("order_date","order_id")
  .rowsBetween(Window.unboundedPreceding, Window.currentRow))
# Unpartitioned running window → Spark WARN WindowExpression
```

Actual Toys ranks: 108=1, 104=2, 102=3. Running ends at 142.50.

WARN: No Partition Defined… Moving all data to a single partition.

Non-fatal here (tiny sample, intentional global cumulative). On large data, partition when the cumulative should restart per key, or accept the cost.

Practice prediction

Toys-only row_number by total desc → 108, 104, 102. No writes in Part 2.

Associate §3 foundation (Aug 2026). Partial coverage. No exam guarantee.
Independent education. Fictional-data project. Part 2 of 3.